

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

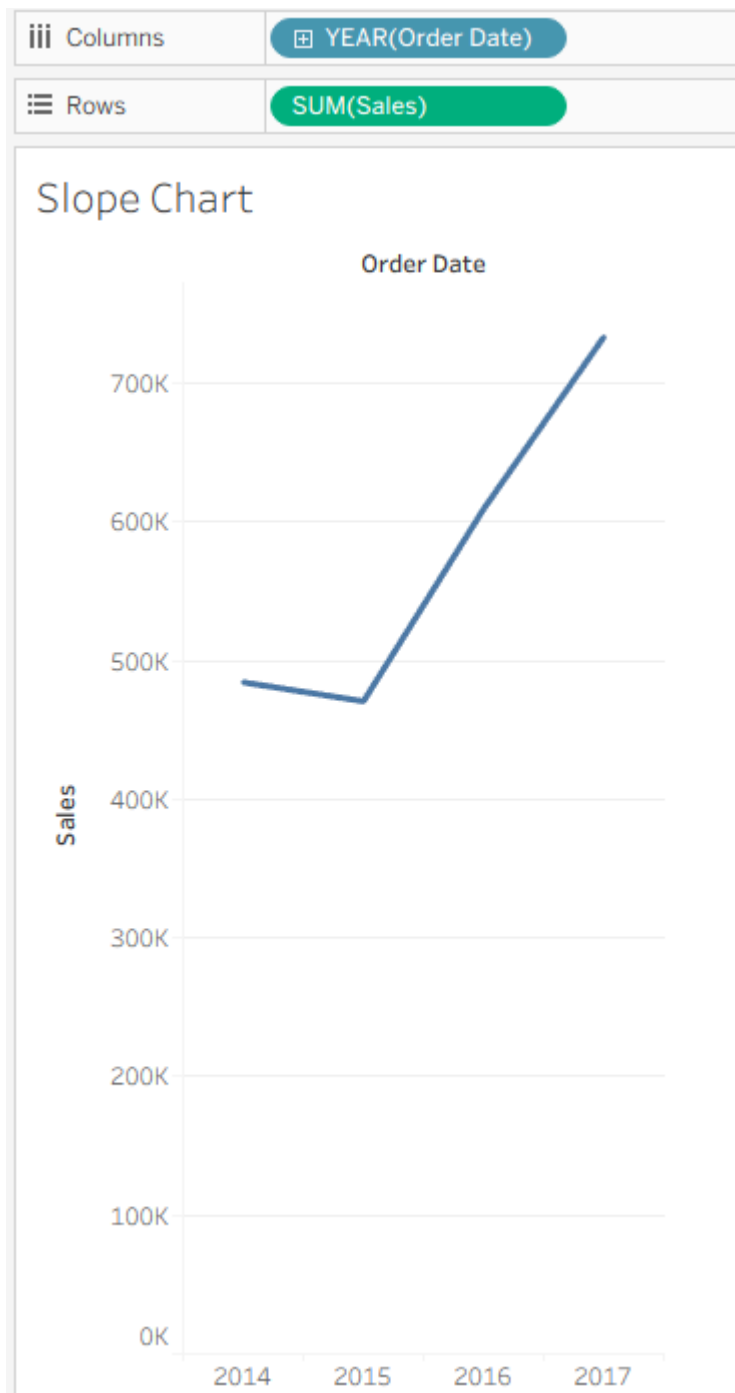
Practical – XI

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Class: TYIT	Batch: 1
Date of Assignment: 01-09-2026	Date/Time of Submission: 01-09-2026

Part B – Program

Implementing Advanced Visualization Techniques and Customizations

- a) Create slope charts and lollipop charts
- Step 1: Set Up the Base Worksheet



Step 2: Filter to Baseline & Comparison Years

Filter Field [Order Date]

How do you want to filter on [Order Date]?

- ☐ Relative Date
- ☐ **Range of Dates**
- ☒ **Years**
- ☐ Quarters
- ☐ Months
- ☐ Days
- ☐ Week numbers
- ☐ Weekdays
- ☐ Month / Year
- ☐ Month / Day / Year
- ☐ Individual Dates

- ☐ Count
- ☐ Count (Distinct)
- ☐ Minimum
- ☐ Maximum

- ☐ Attribute

Next > Cancel

Filter [Year of Order Date]

General Condition Top

☒ Select from list ☐ Custom value

Enter search text

- ☒ 2014
- ☐ 2015
- ☐ 2016
- ☒ 2017

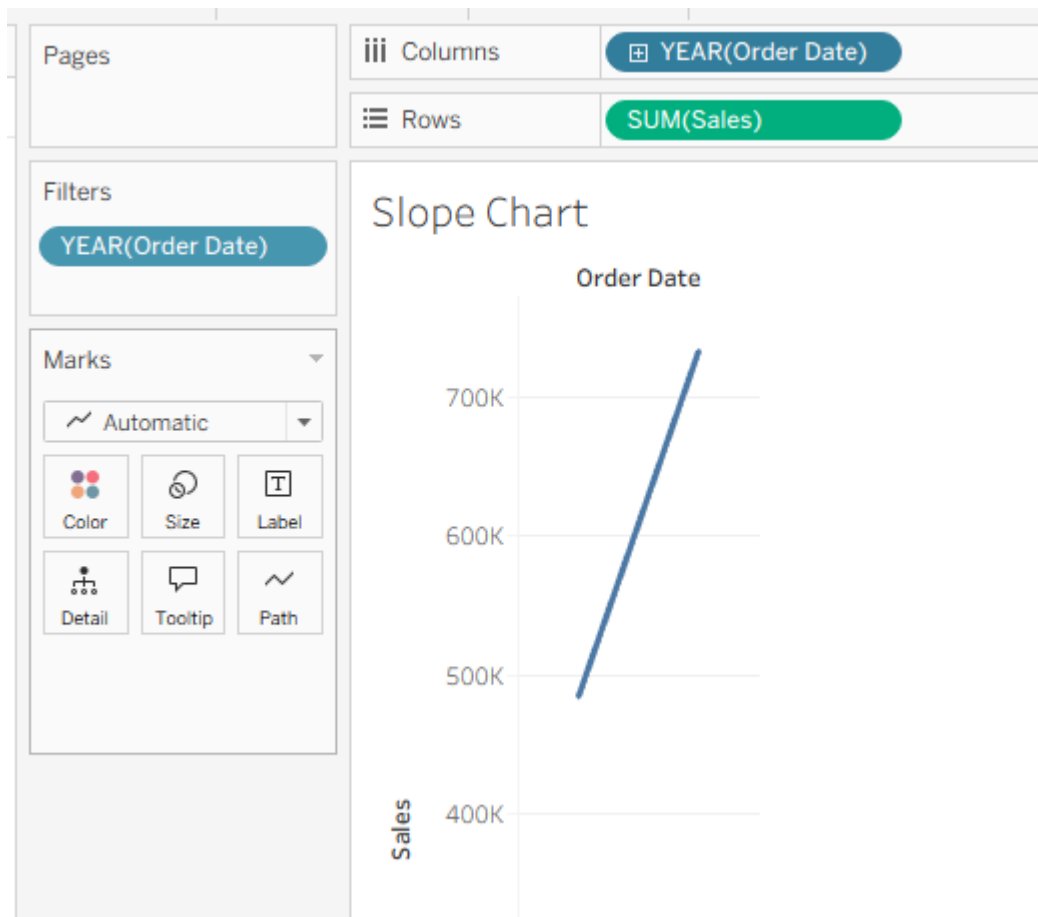
All None

Summary

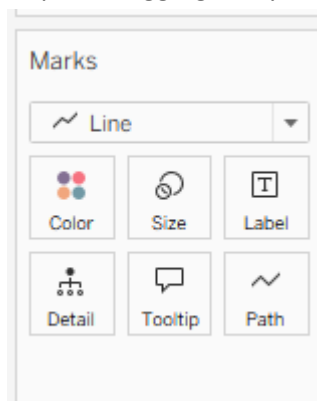


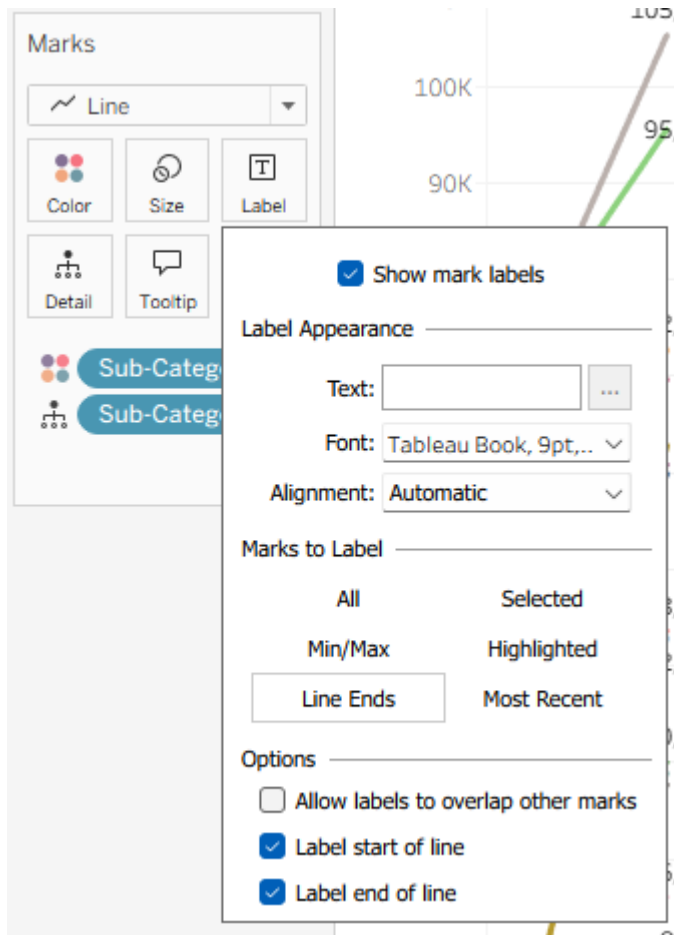
The screenshot shows the Power BI Desktop interface. On the left, a line chart is partially visible with a blue line. The main area displays the 'Format' pane for the 'YEAR(Order Date)' field. The 'Time Granularity' section is expanded, showing a list of options: Year, Quarter, Month, Day, and More. The 'Discrete' option is selected, highlighted in blue. Other options like 'Continuous' are visible below. The 'Attribute' and 'Measure' sections are also visible but not expanded.

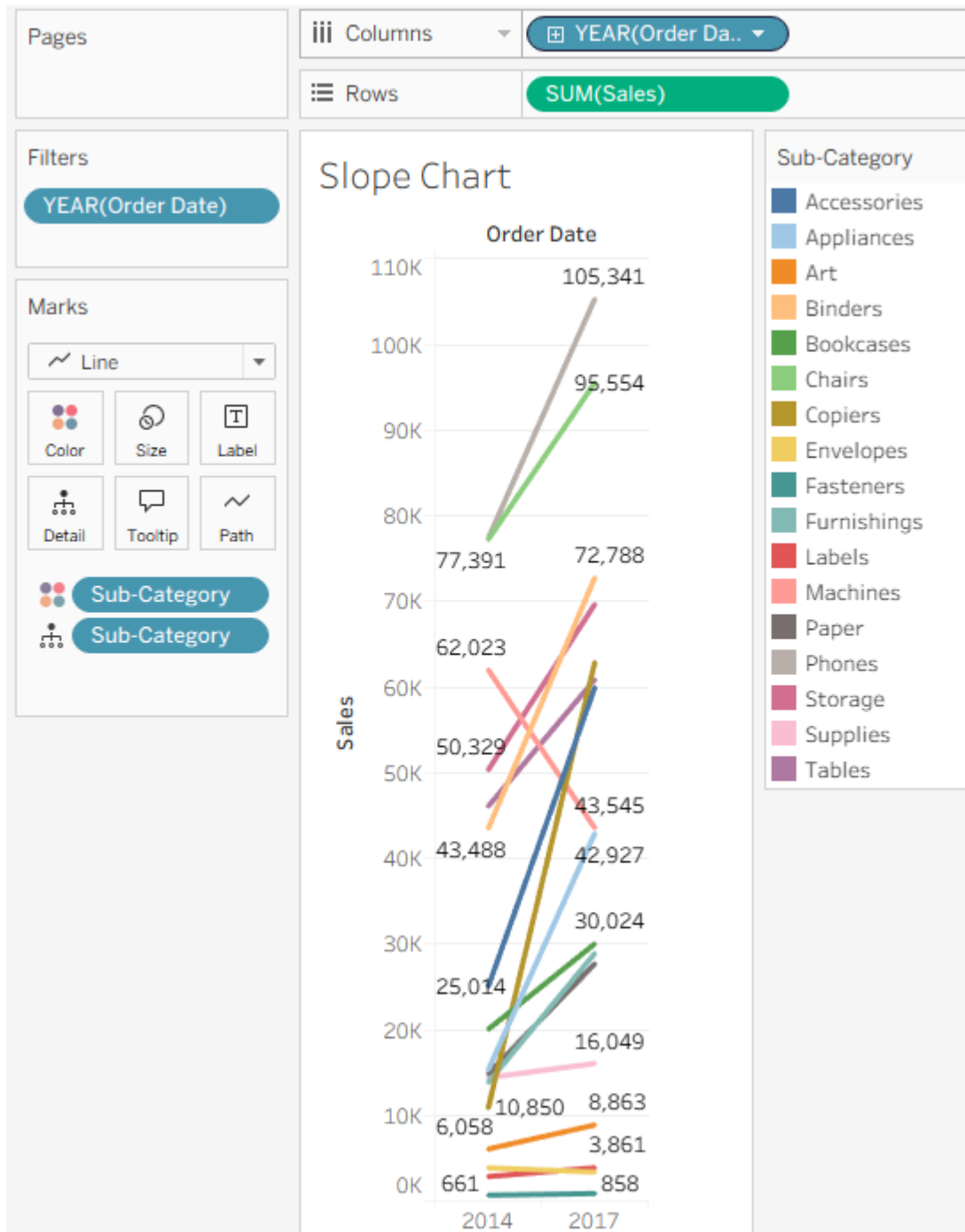
Field	Time Granularity	Attribute	Measure
YEAR(Order Date)	Discrete		



Step 3: Disaggregate by Sub-Category







Slope Chart

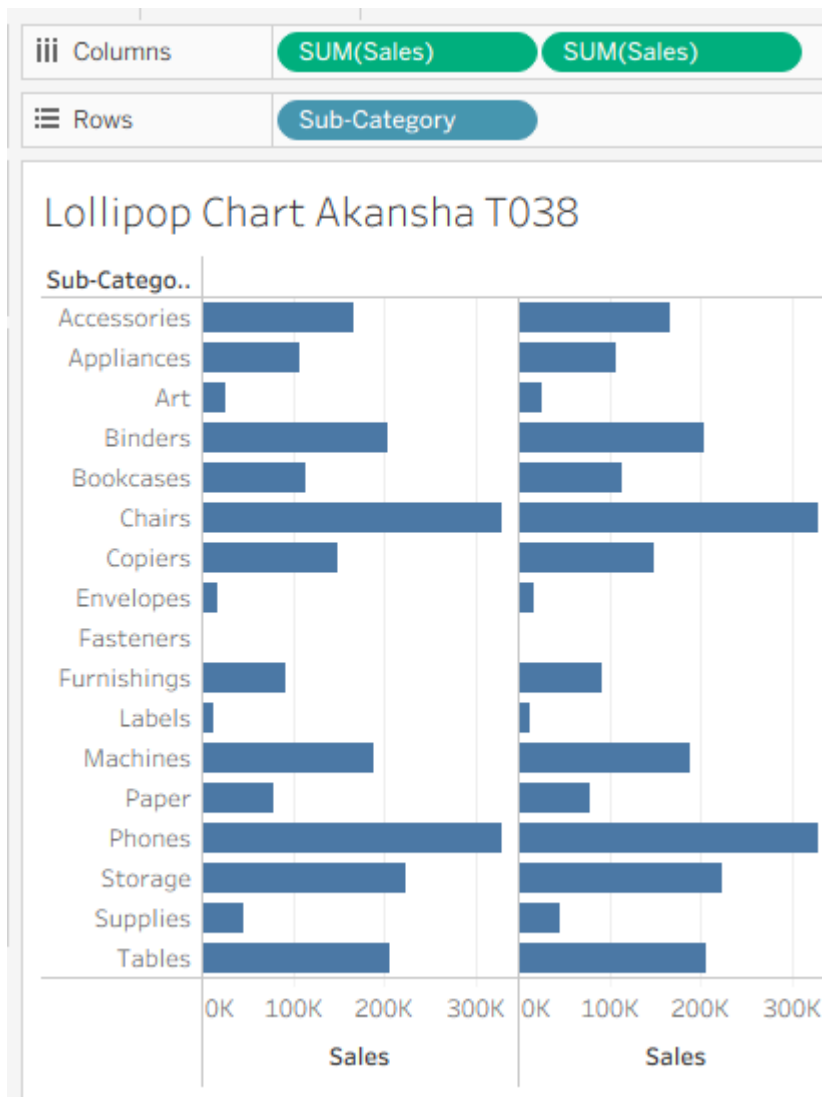
The chart illustrates the sales trend for 17 different sub-categories from 2014 to 2017. The y-axis represents Sales in thousands (K), ranging from 0K to 110K. The x-axis shows the years 2014 and 2017. Each line represents a sub-category, with the 2017 sales value labeled at the end of the line.

Sub-Category	2014 Sales (K)	2017 Sales (K)
Accessories	43,488	72,788
Appliances	42,927	105,341
Art	6,058	3,861
Binders	10,850	8,863
Bookcases	25,014	16,049
Chairs	77,391	95,554
Copiers	43,488	62,023
Envelopes	6,058	3,861
Fasteners	661	858
Furnishings	10,850	8,863
Labels	50,329	43,545
Machines	62,023	72,788
Paper	25,014	16,049
Phones	43,488	72,788
Storage	50,329	43,545
Supplies	10,850	8,863
Tables	661	858

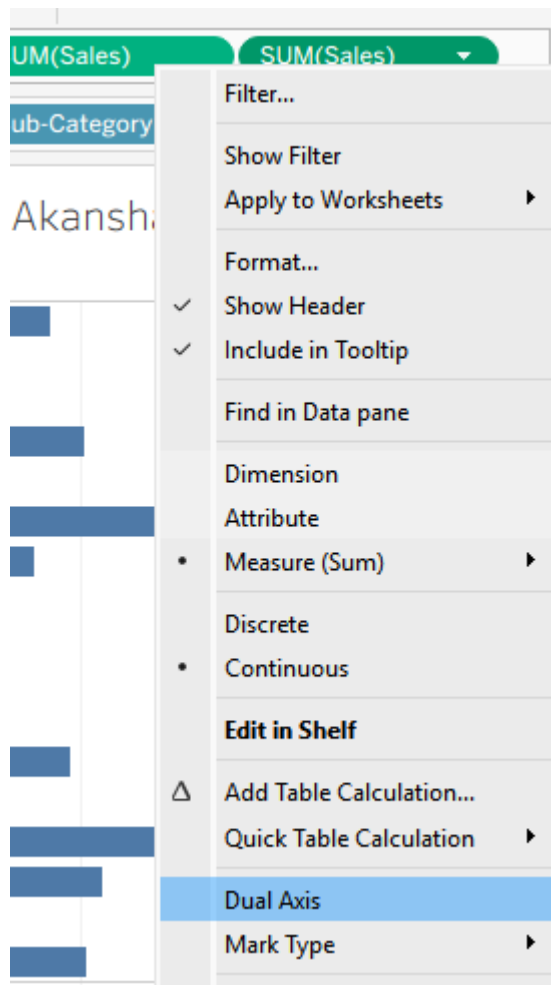


2. Create a Lollipop Chart (Sales with Profit Highlighting)

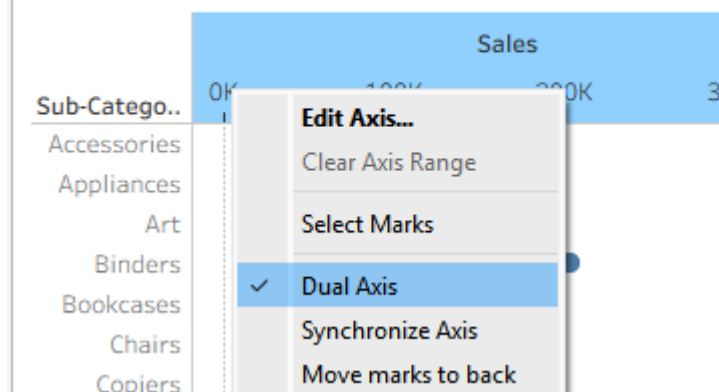
Step 1: Set Up the Base Worksheet

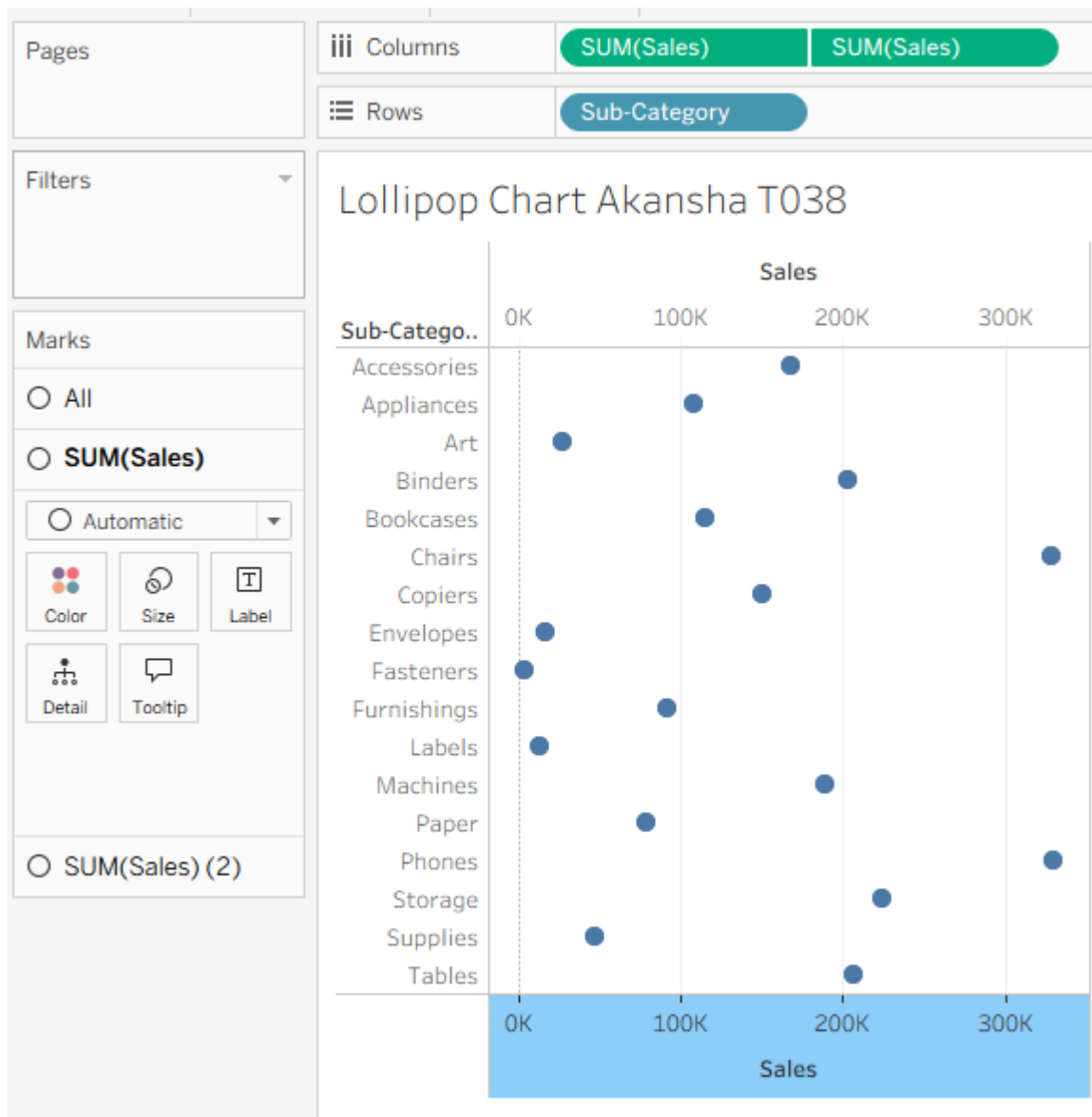


Step 2: Convert to Dual Axis

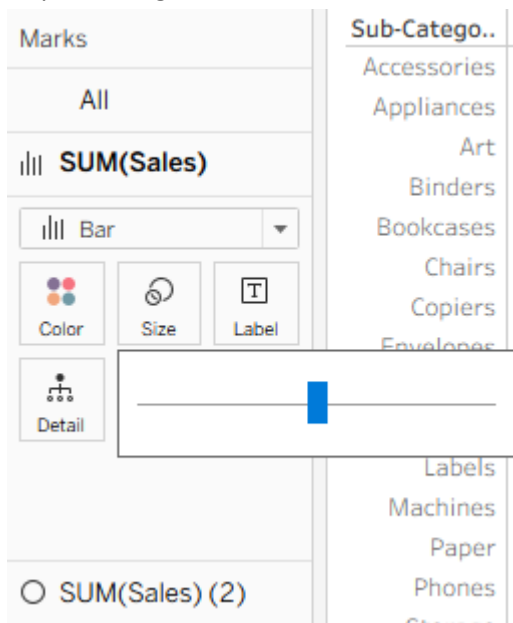


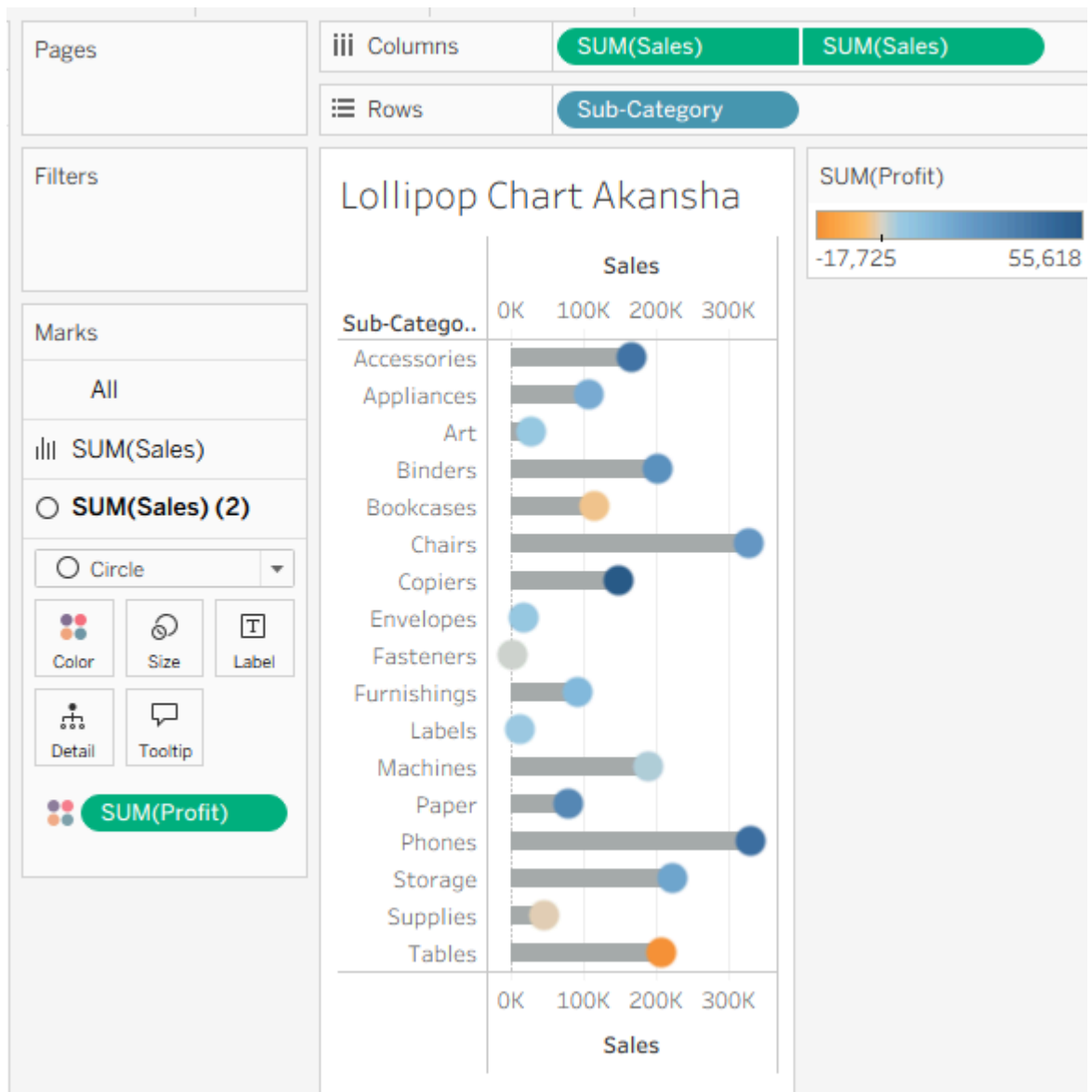
Lollipop Chart Akansha T038



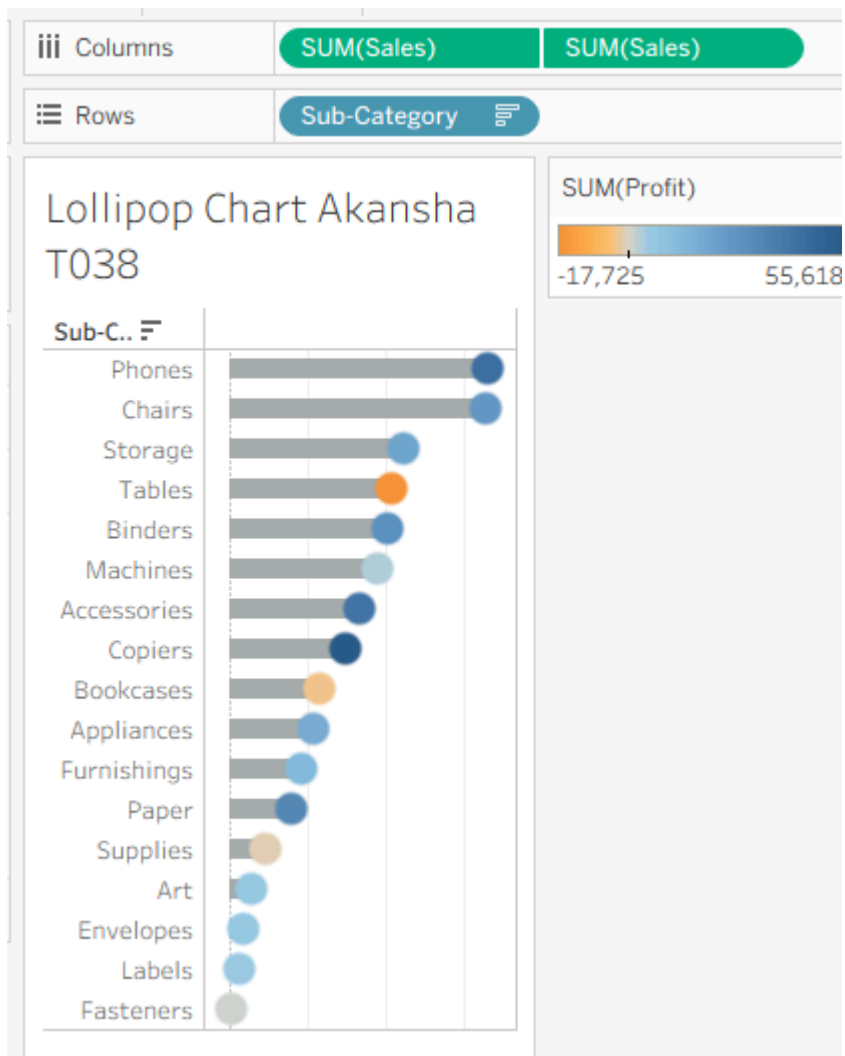


Step 3: Configure the Marks Cards



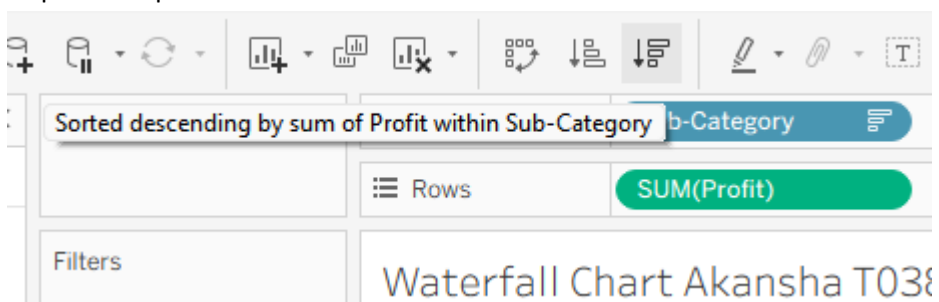


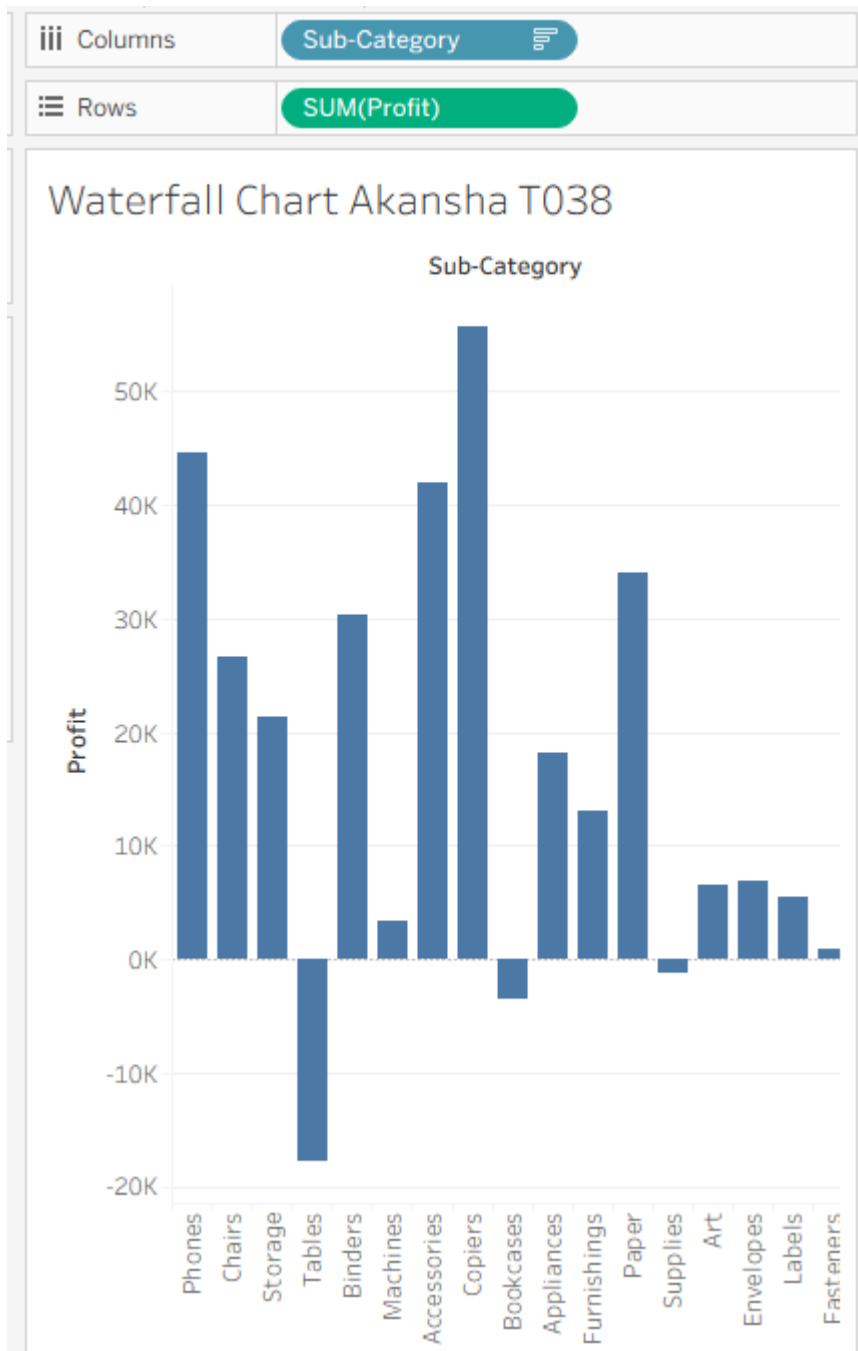
Step 4: Clean Up Axis Formatting



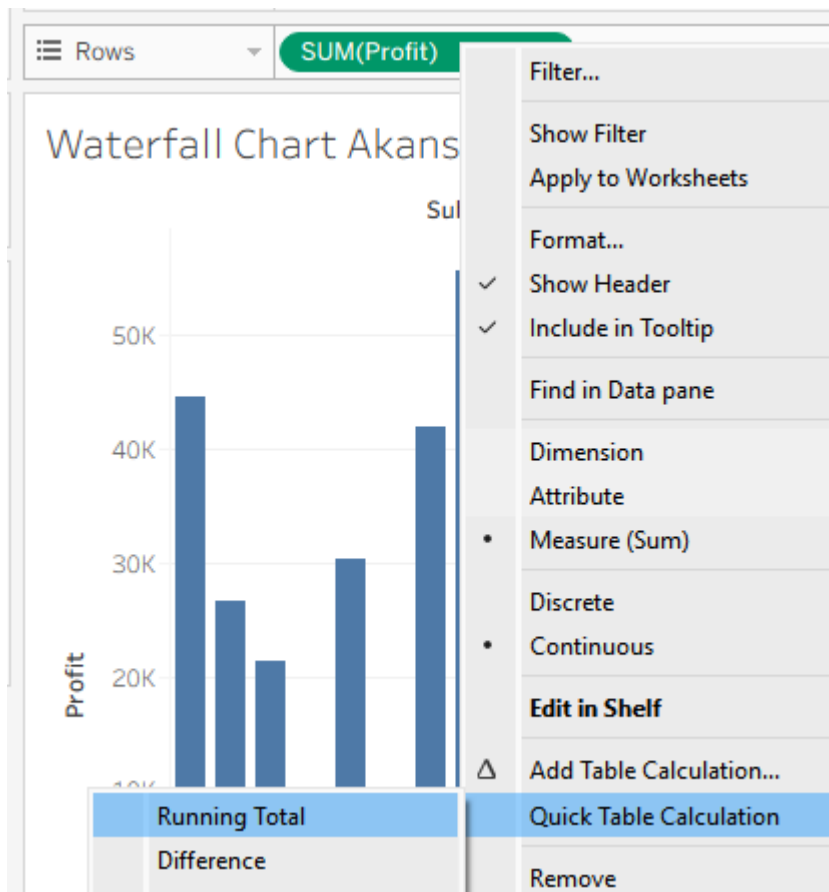
b) Design waterfall charts

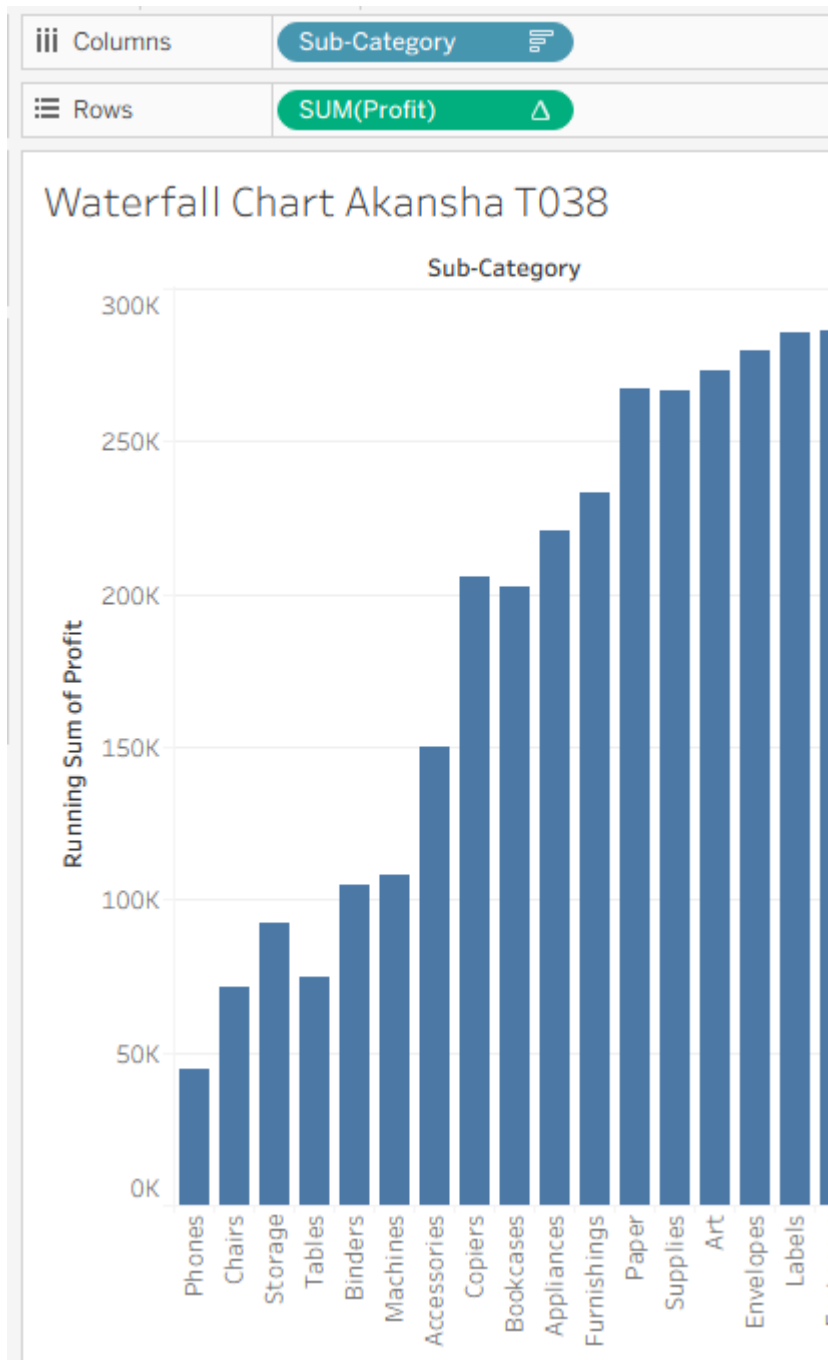
Step 1: Set Up the Base Worksheet



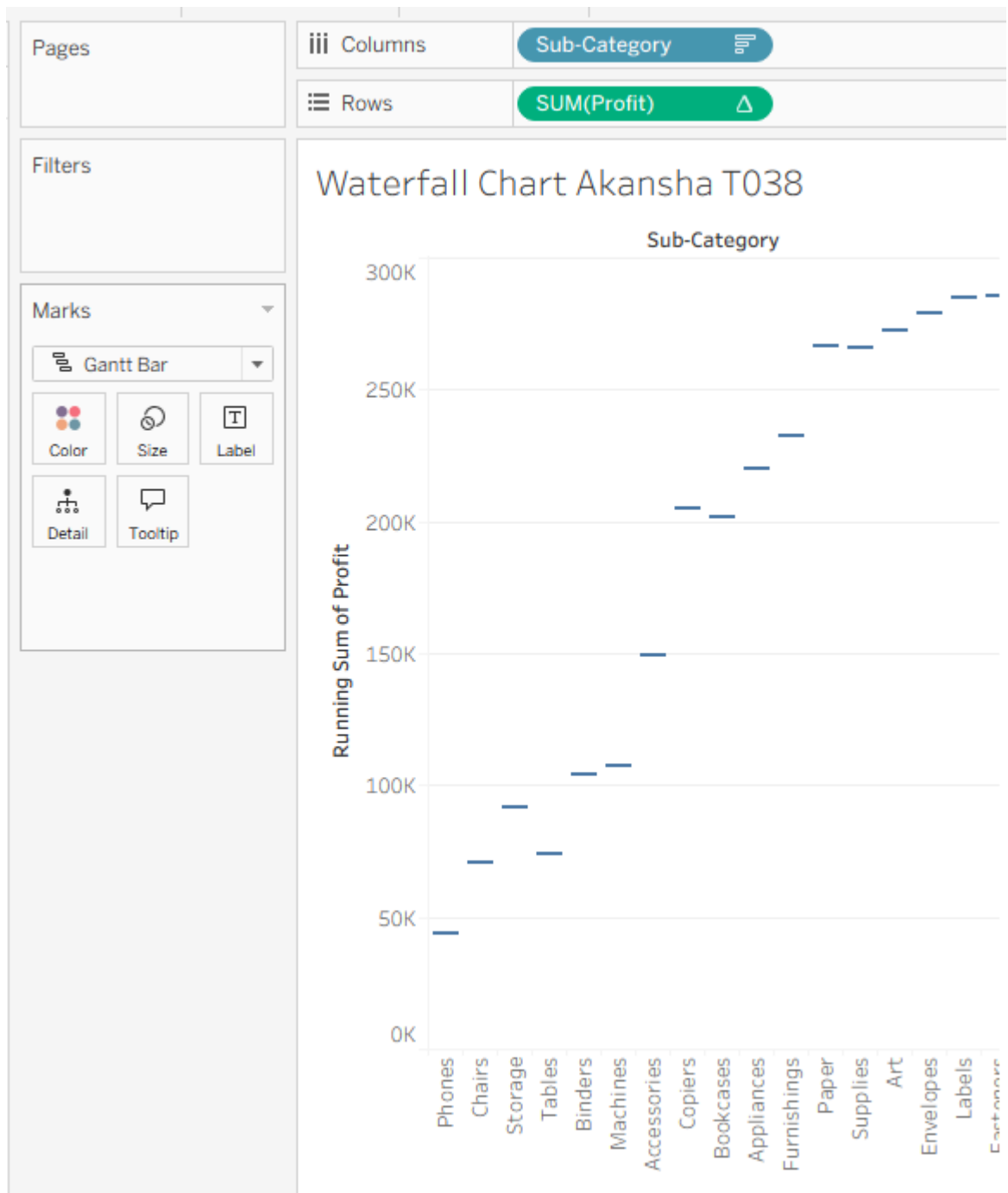


Step 2: Apply Running Total Table Calculation

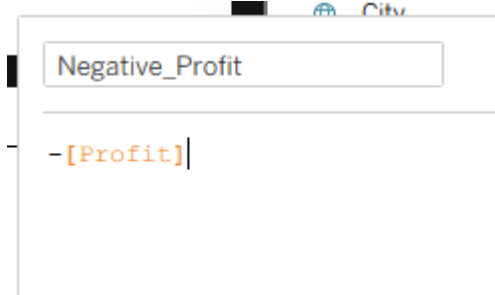




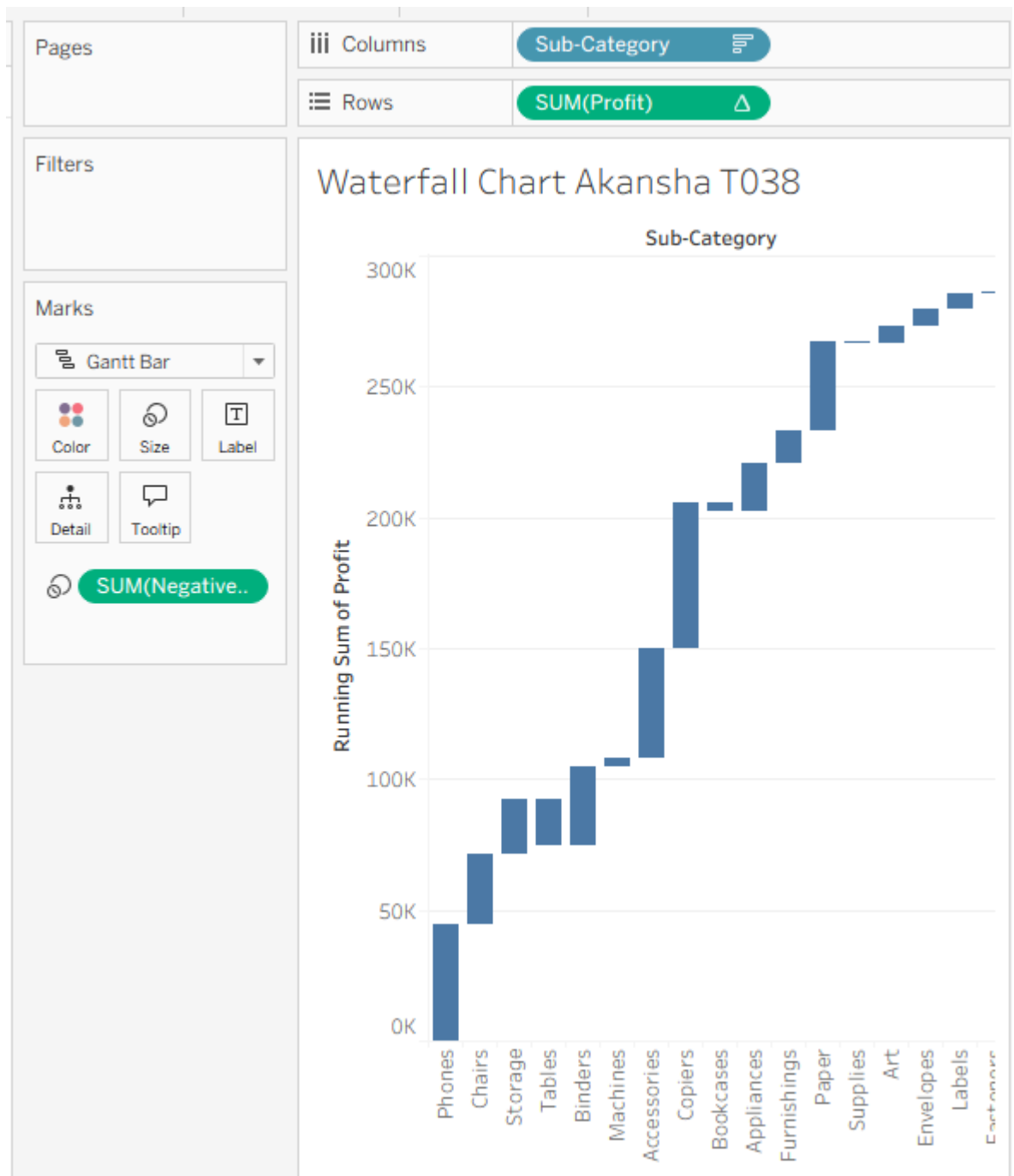
Step 3: Switch Mark Type to Gantt Bar



Step 4: Create the Negative Profit Sizing Calculation



Step 5: Size and Color the Waterfall Blocks



Palette

Automatic	
	Red-Gold
	Orange-Blue Diverging
	Red-Green Diverging
	Green-Blue Diverging
	Red-Blue Diverging

Edit Colors SUM(Profit)

Palette

Red-Green Diverging

-17,725 55,618

Start

Automatic -17,725.4811

Center

Automatic 0

End

Automatic 55,617.8249

☐ Stepped Color 5 steps

☐ Reverse

☒ Use full color range

☐ Include totals

Step 6: Add Grand Totals to Close the Waterfall

Worksheet Dashboard Story Analysis Map Format Server Window Help

← → ↺ 🏠 + 📄

Analytics < Pages

Sample - Superstore

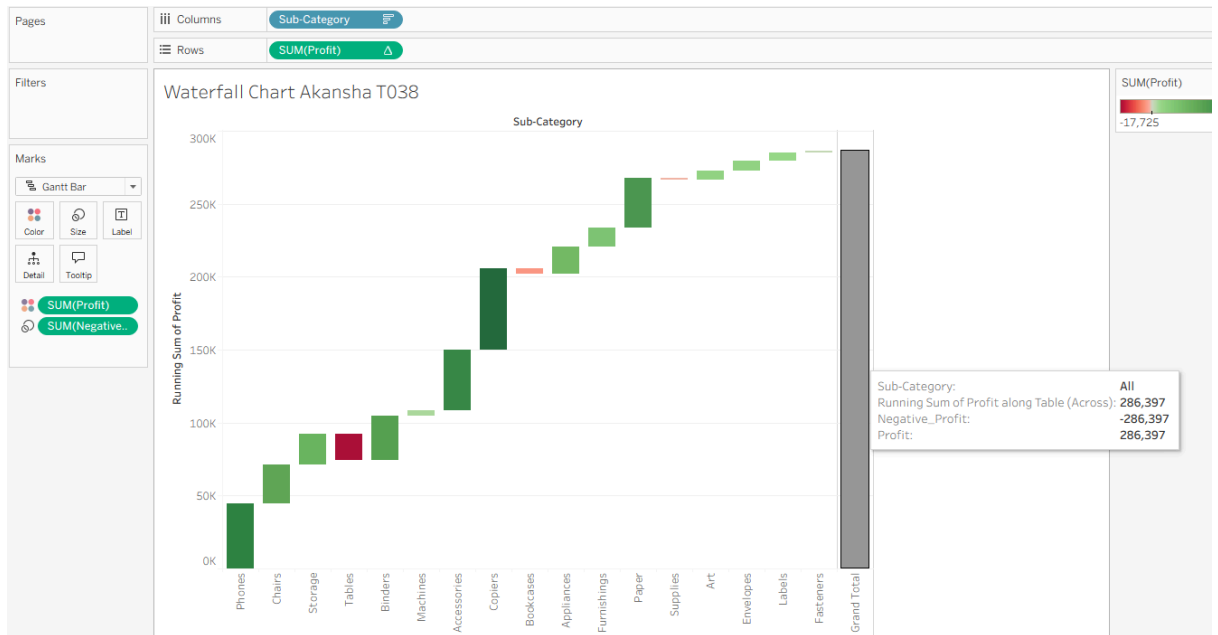
🔍 🔍 📄

Filters

Category

- Show Row Grand Totals
- Show Column Grand Totals
- Row Totals to Left
- Column Totals to Top
- Add All Subtotals
- Remove All Subtotals

- Show Mark Labels
- ✓ Aggregate Measures
- Stack Marks
- View Data...
- Explain Data Settings...
- Reveal Hidden Data
- Percentage Of
- Totals
- Forecast
- Trend Lines
- Special Values
- Table Layout
- Legends



c) Build sparklines for trend analysis

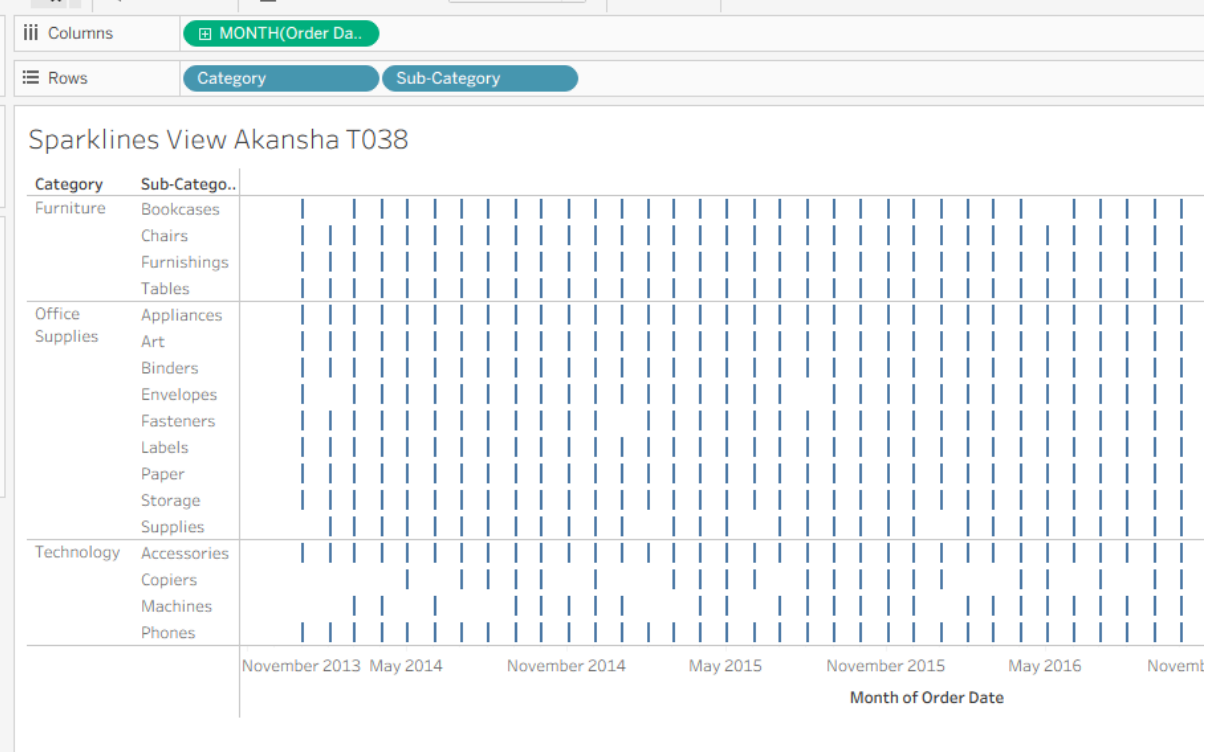
Step 1: Set Up the Base Worksheet and Table Dimensions

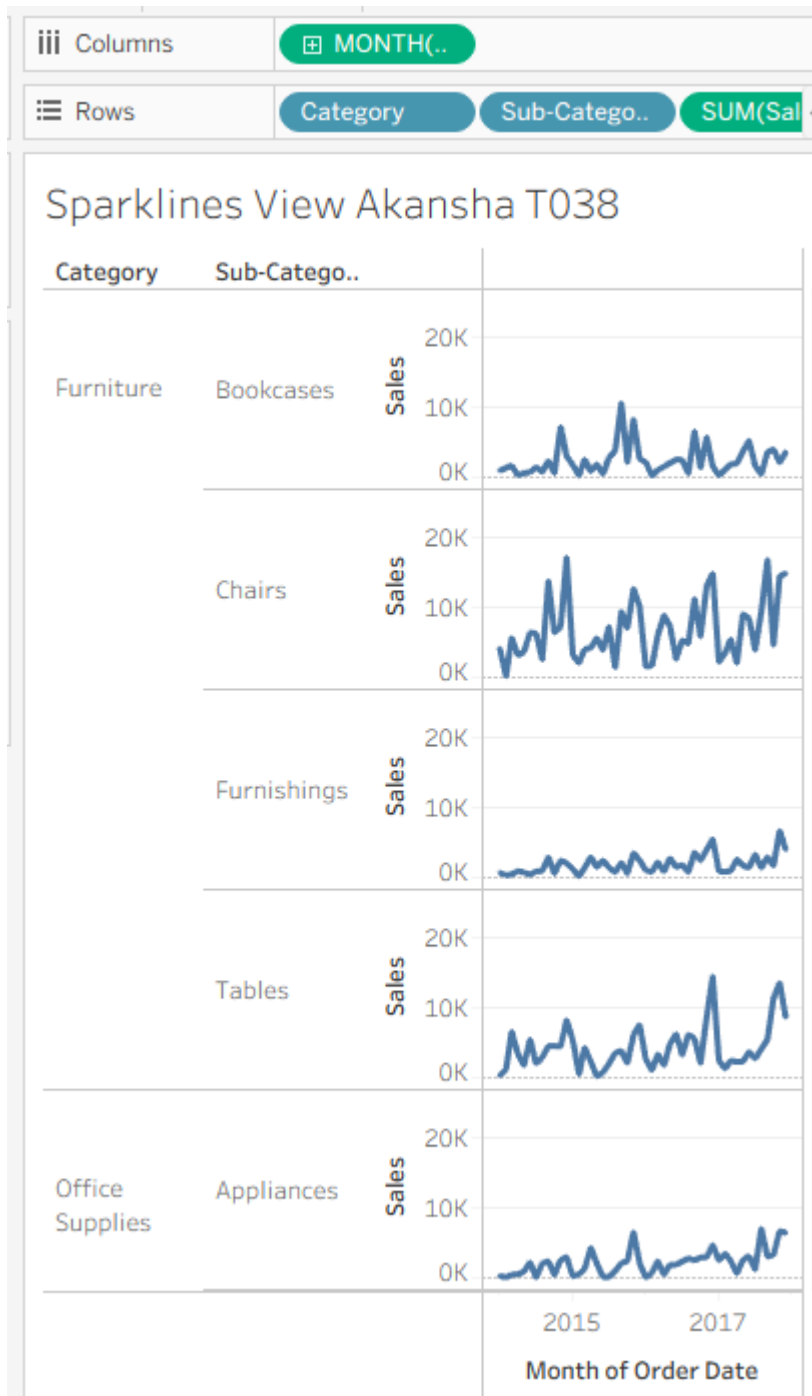
Sparklines View Akansha T038

Category	Sub-Catego..	
Furniture	Bookcases	Abc
	Chairs	Abc
	Furnishings	Abc
	Tables	Abc

Step 2: Add Continuous Time-Series for Trend Shape

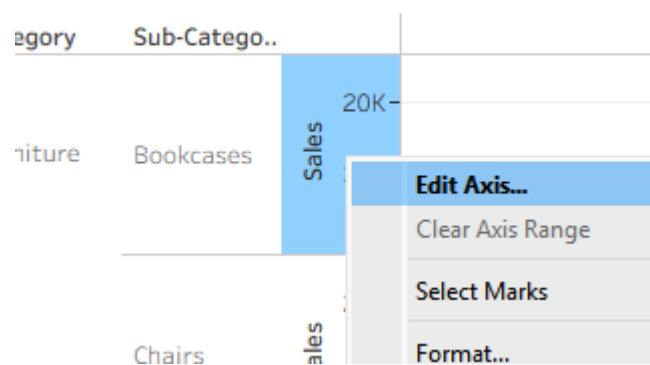
Year	2015
Quarter	Q2 2015
• Month	May 2015
Week Number	Week 5, 2015
Day	May 8, 2015
More	►
Exact Date	
Attribute	
Measure	►
Discrete	
• Continuous	





Step 3: Add Total Sales Metric Pill alongside the Sparkline

arklines View Akansha T038



Edit Axis [Sales]

General

Tick Marks

Range

☐ Automatic
☒ Include zero

☐ Uniform axis range for all rows or columns

☒ Independent axis ranges for each row or column

☐ Custom

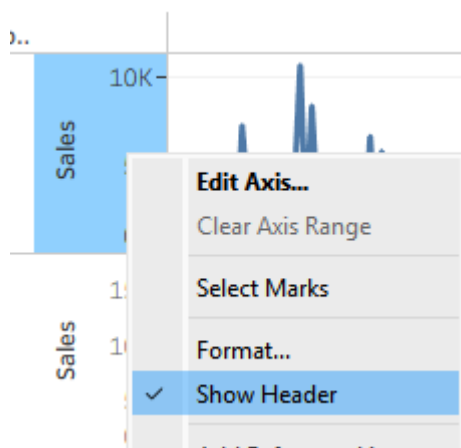
Independent

Independent

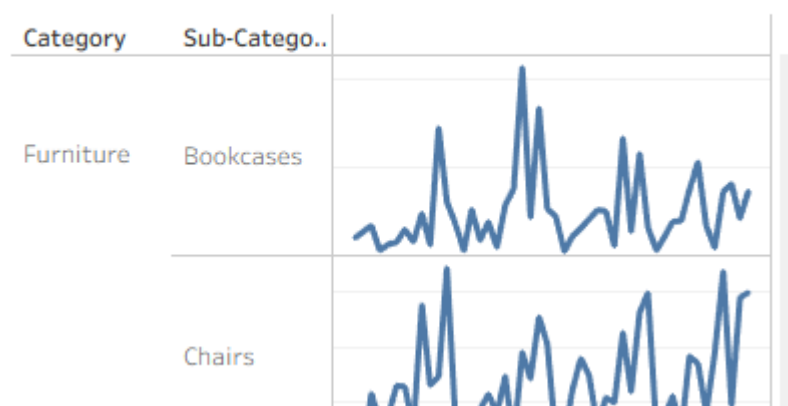
-1.433.1298176

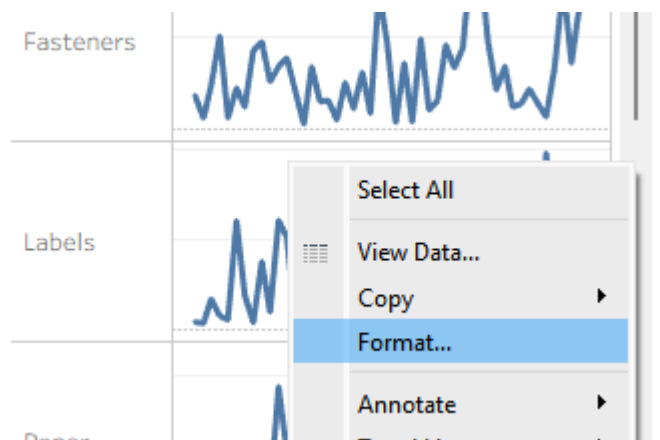
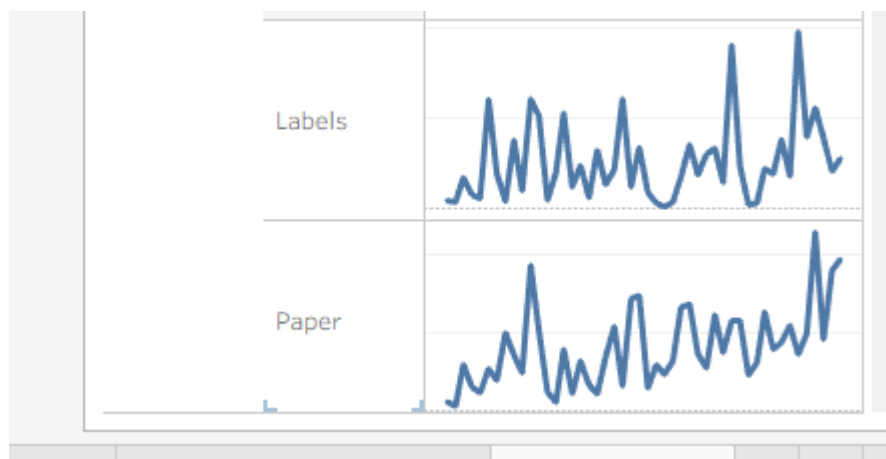
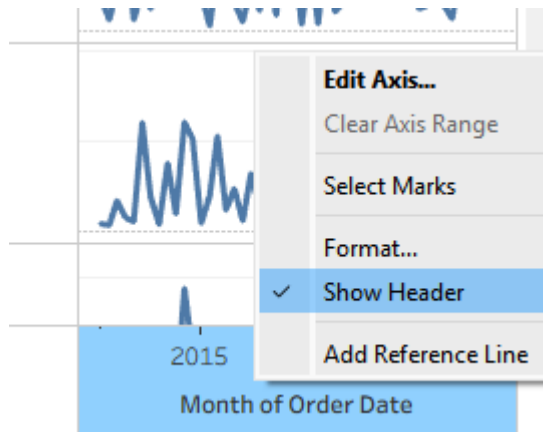
26.994.4707615

Step 4: Strip Away Visual Clutter (Clean Sparkline Design)



Sparklines View Akansha T038





iii Columns

⊕ MONTH(Order D..

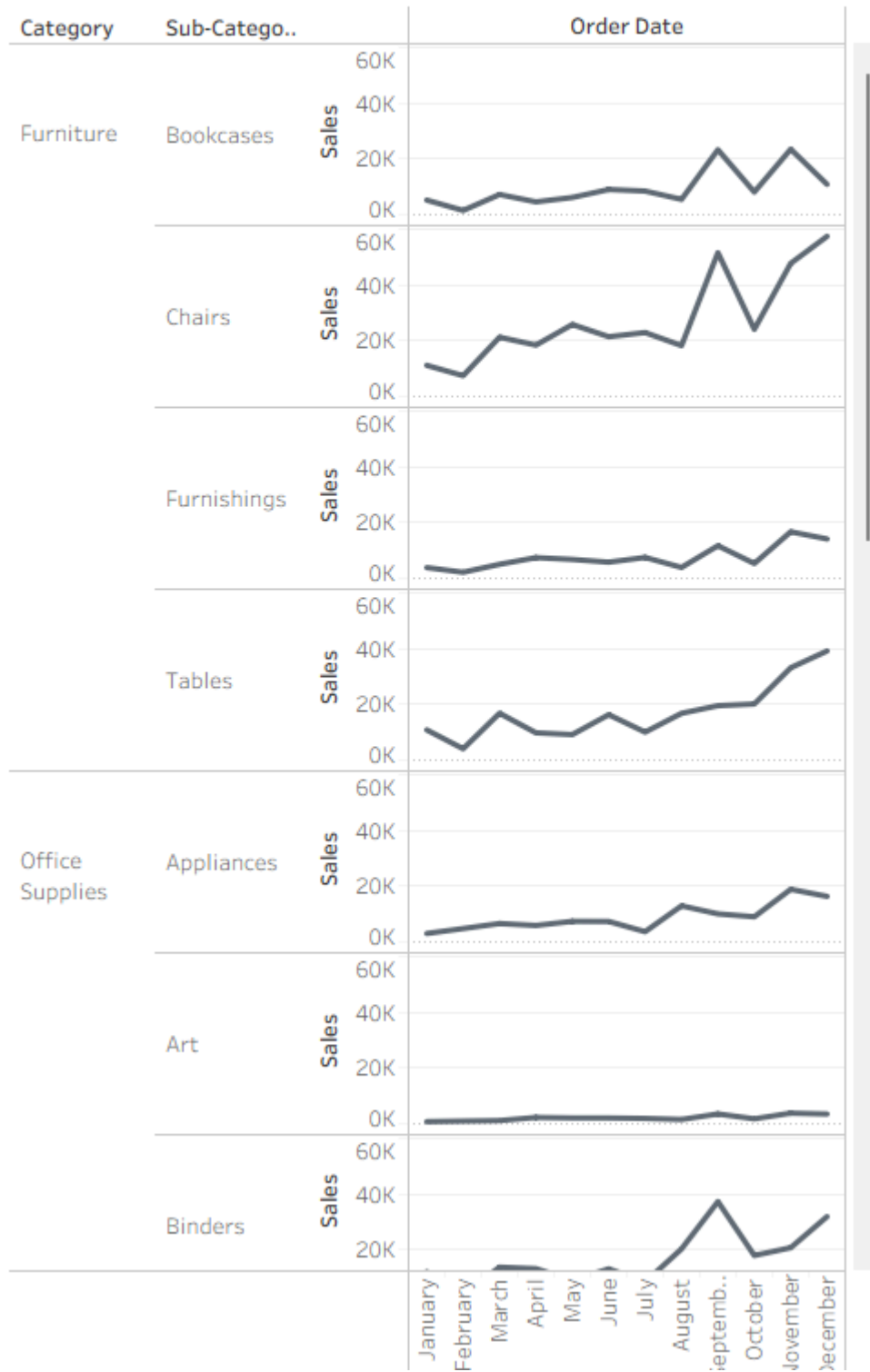
Rows

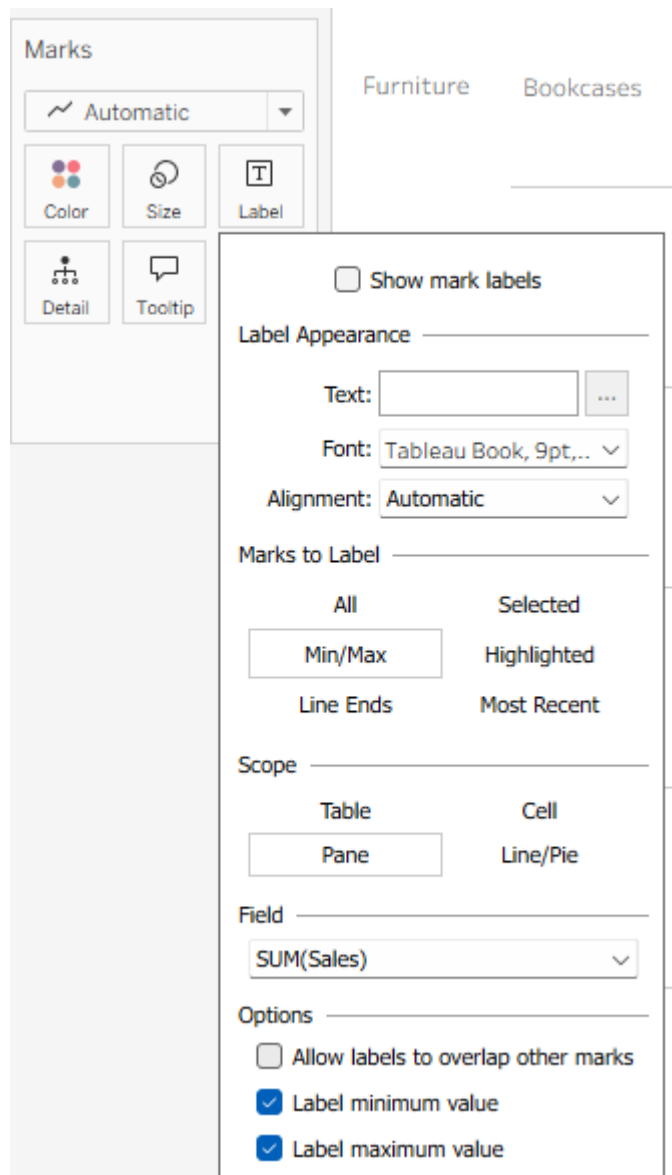
Category

Sub-Category

SUM(Sales)

Sheet 1





d) Implement sheet swapping for dynamic dashboards
Step 1: Create the Control Parameter

Create Parameter

×

Name

Select_View_Parameter

Properties

Data type

String

▼

Display format

Category Sales (Bar)

▼

Current value

Category Sales (Bar)

▼

Value when workbook opens

Current value

▼

Allowable values

☐ All

☒ List

☐ Range

Value	Display As	
Category Sales (Bar)	Category Sales (Bar)	
Regional Profit (Map)	Regional Profit (Map)	
Click to add		

☒ Fixed

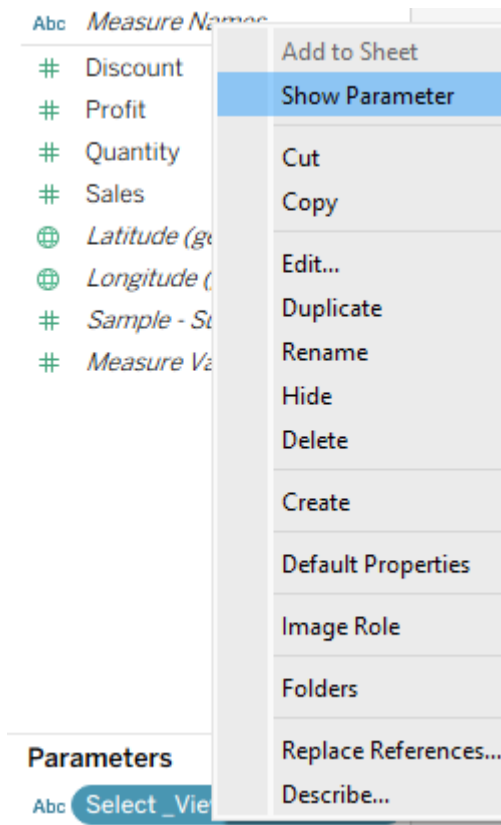
☐ When workbook opens

Add values from ▼

Remove Selected

Cancel

OK

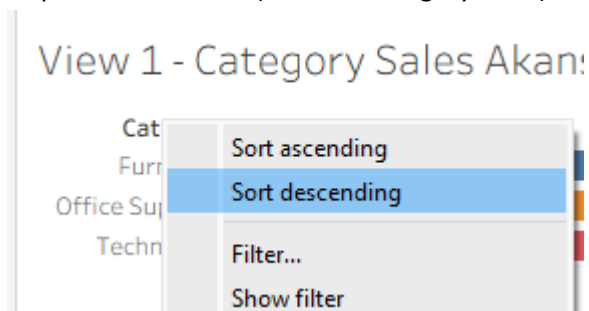


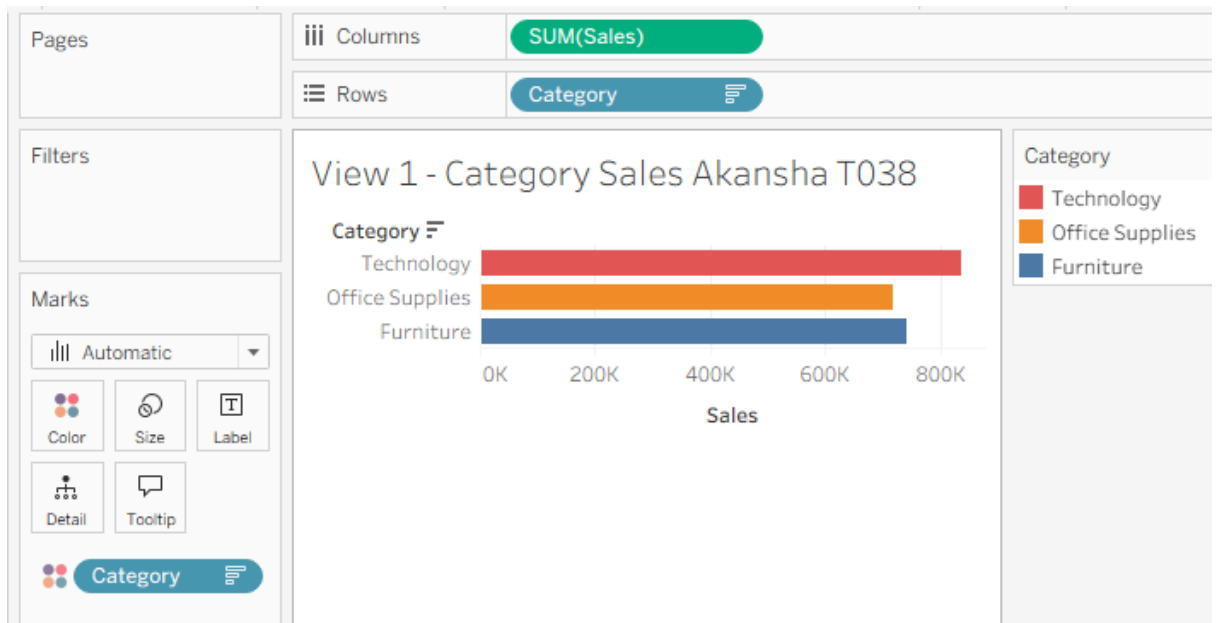
Step 2: Create the Dynamic Filter Field

View_Filter

[Select_View_Parameter]

Step 3: Build Sheet 1 (View 1 - Category Sales)





Filter [View_Filter]

General Wildcard Condition Top

☐ Select from list ☒ Custom value list ☐ Use all

Category Sales (Bar)

Clear List ☒ Include all values when empty ☐ Exclude

Summary

Field: [View_Filter]
 Selection: Selected all values
 Wildcard: All
 Condition: None
 Limit: None

Reset OK Cancel Apply

Step 4: Build Sheet 2 (View 2 - Regional Profit)
 Drag the state column or double-click on it

Edit Parameter [Select_View_Parameter]

Name

Select_View2_Parameter

Properties

Data type

String

Display format

Regional Profit (Map)

Current value

Regional Profit (Map)

Value when workbook opens

Current value

Allowable values

☐ All

☒ List

☐ Range

Value	Display As	
Category Sales (Bar)	Category Sales (Bar)	
Regional Profit (Map)	Regional Profit (Map)	
Click to add		

☒ Fixed

☐ When workbook opens

Add values from

Remove Selected

Cancel

OK

Filter [View_Filter]

General

Wildcard

Condition

Top

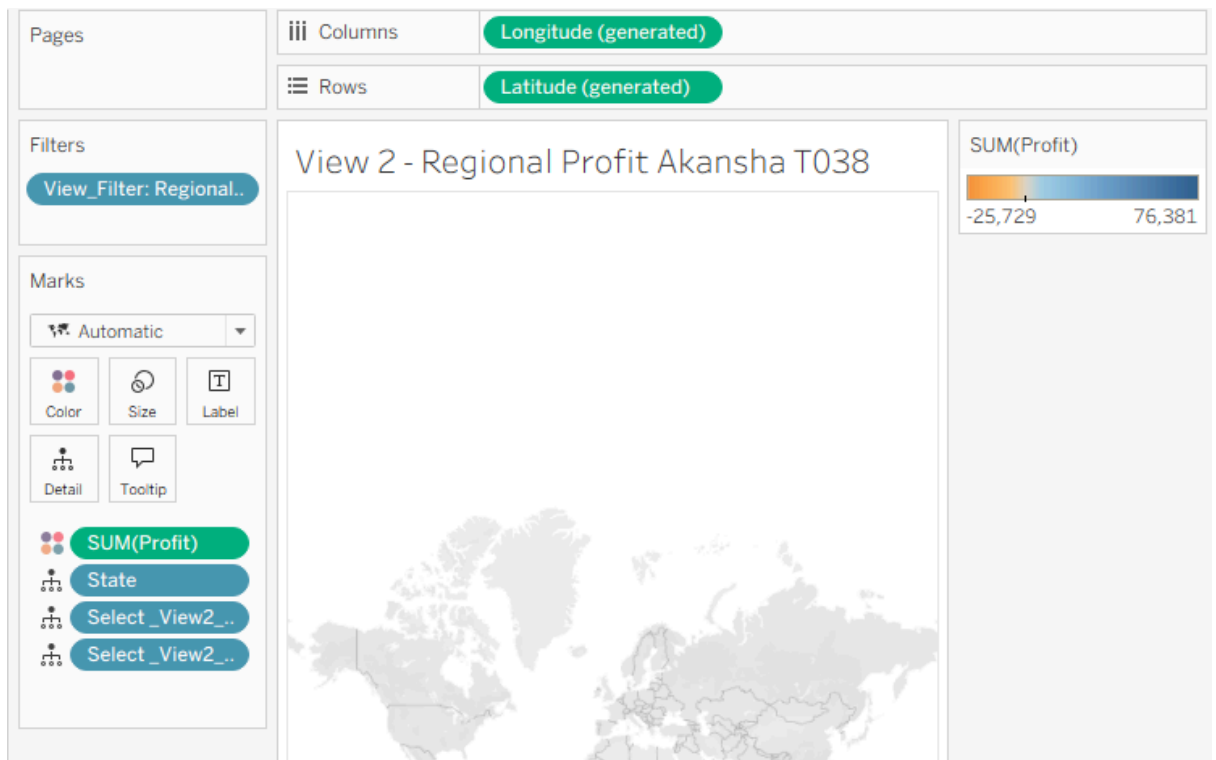
☒ Select from list

☐ Custom value list

☐ Use all

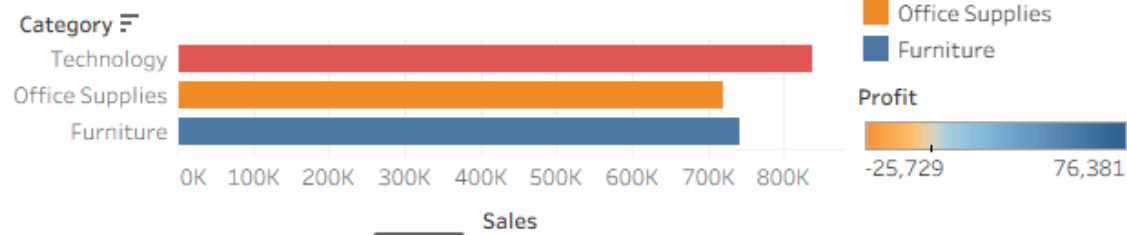
Enter search text

☒ Regional Profit (Map)



Step 5: Construct the Dynamic Swapping Dashboard

View 1 - Category Sales Akansha T038

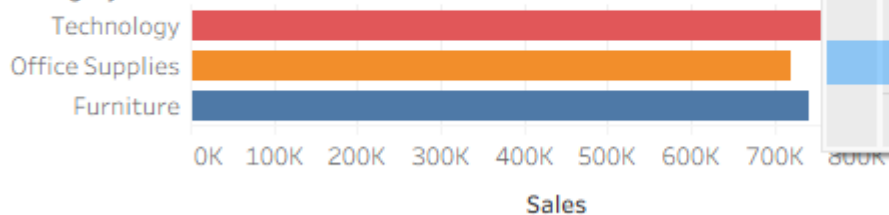


View 2 - Regional Profit Akansha T038



View 1 - Category Sales Akansha T038

Category



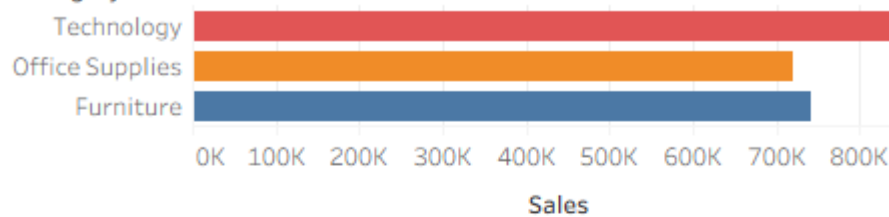
Edit Title...

Reset Title

Hide Title

Format Title...

Category



Category

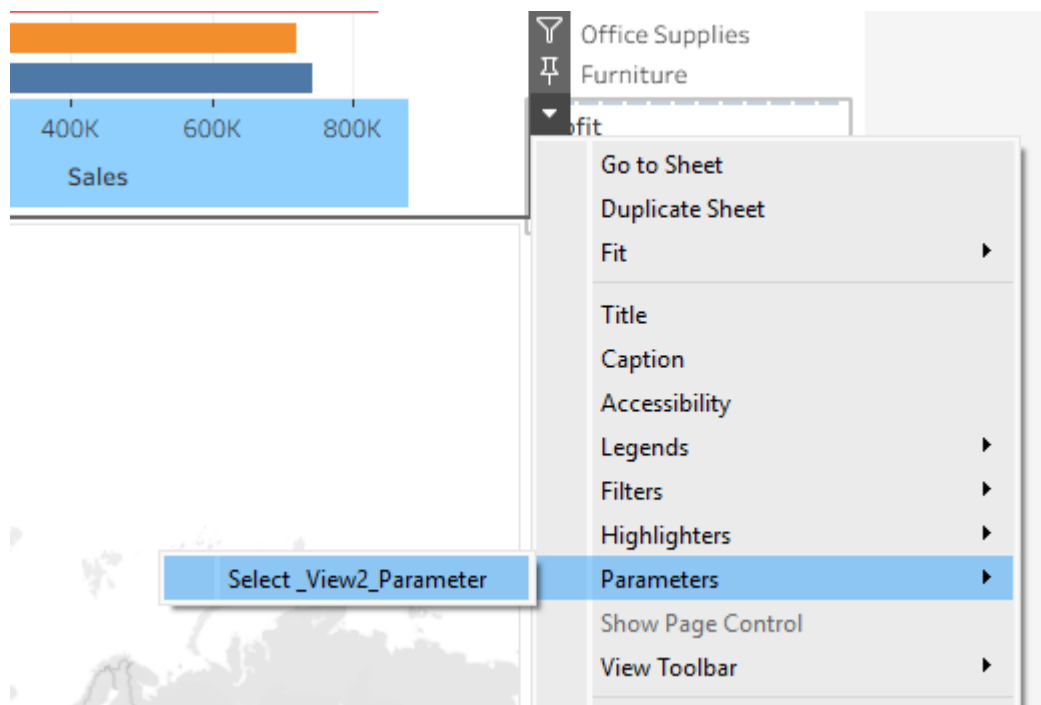
Technology

Office Supplies

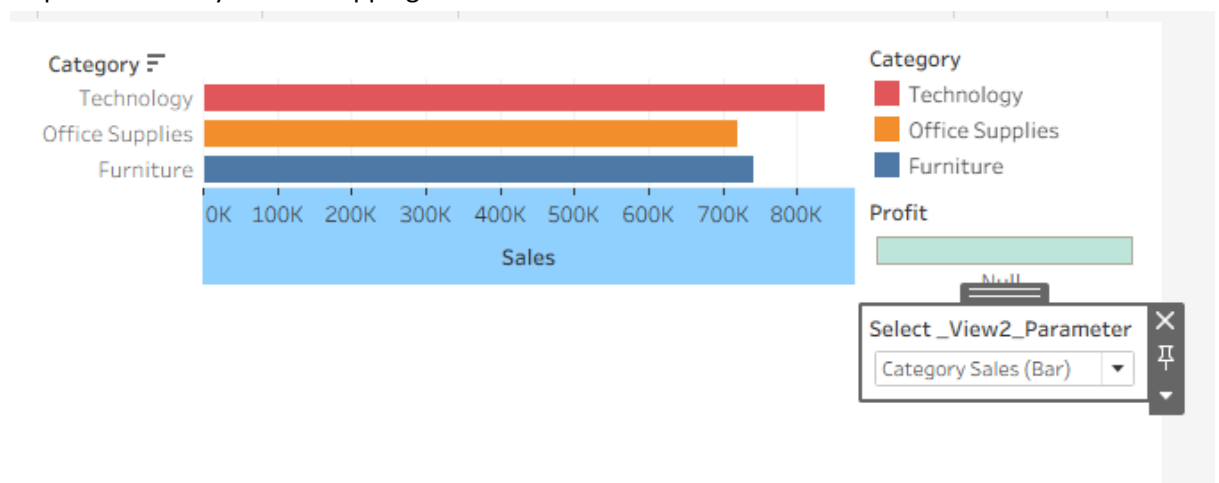
Furniture

Profit

-25,7%

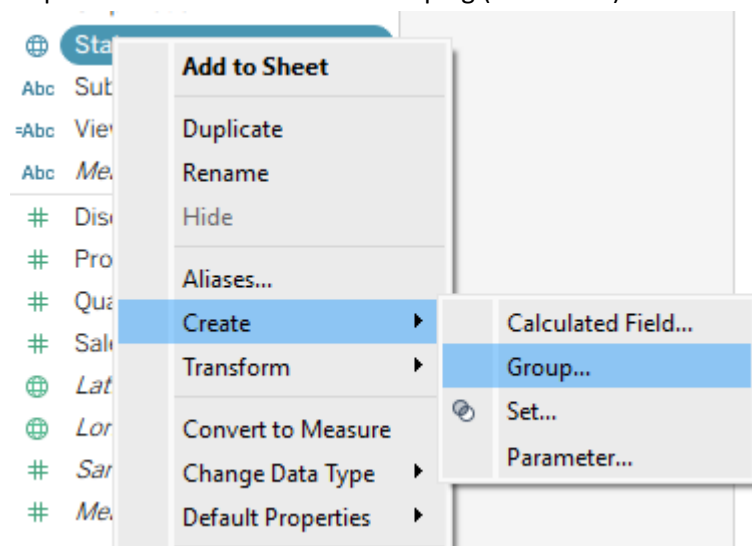


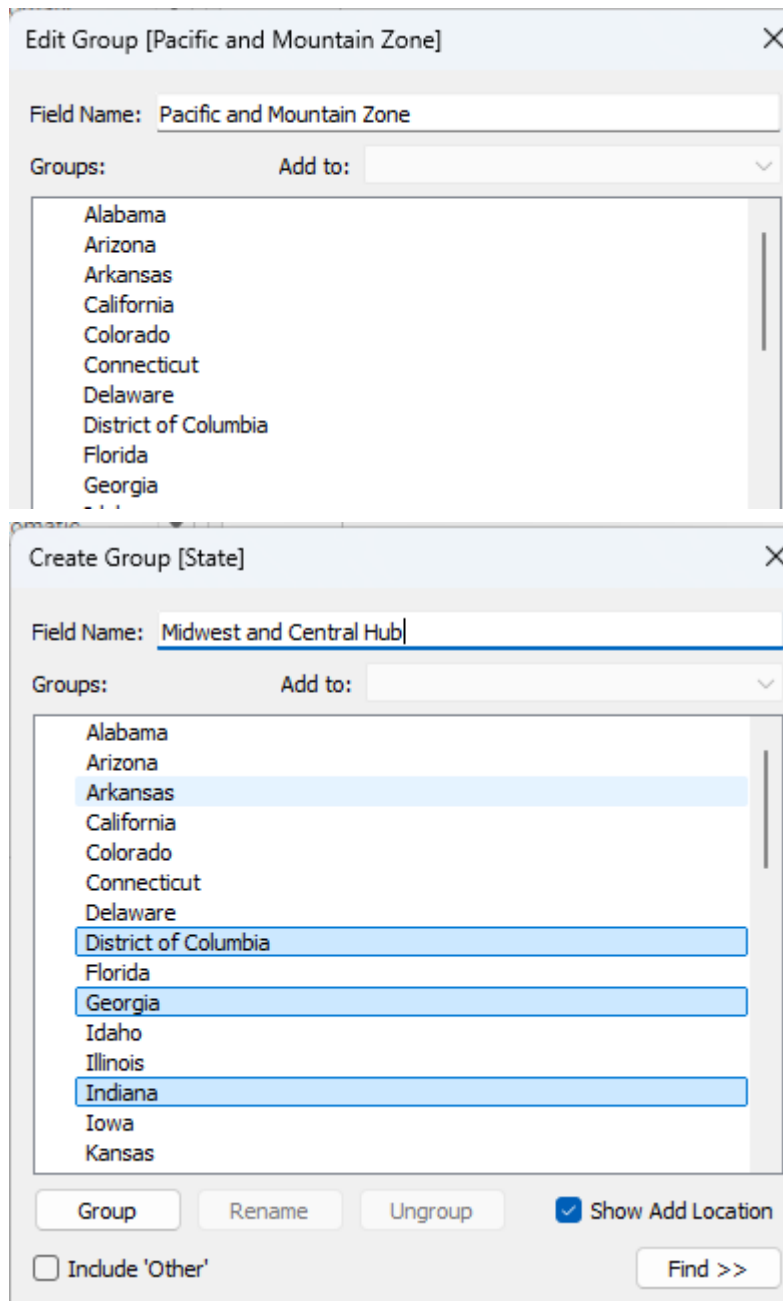
Step 6: Test the Dynamic Swapping



e) Create custom geographic territories

Step 1: Create a Custom State Grouping (Territories)





select any city name

Step 2: Assign Geographic Role to the Custom Group

f) Use background images in dashboards

g) Apply animation and transparency techniques

Part C – Curiosity Question

1. What is a slope chart? Draw a simple slope chart showing a change between two years.
2. What is a lollipop chart? Draw a simple lollipop chart for comparing three categories.
3. What is a waterfall chart? Draw a simple waterfall chart showing increase and decrease in values.

4. What is sheet swapping in Tableau? Draw a simple diagram showing how two worksheets can be switched using a parameter.
5. What is a sparkline? Draw a simple sparkline showing a sales trend over time.